Ubiquiti Networks Deployment Process

Thursday, November 2, 2023 1:31 PM

Naming Conventions

Thursday, November 2, 2023 1:31 PM

Naming convention:

Use proper templated names for the primary externally-visible devices such as UDMs UDM Pros UDRs, Dream Walls, USGs, or Enterprise Gateways:

[COMPANY CODE]-[DEVICE CODE]-[TWO LETTER STATE ABBREVIATION]-[CITY ABBREVIATION]

Valid device codes:

[UDM] - Dream Machine
[UDMPRO] - Dream Machine Pro
[UDMSE] - Dream Machine SE
[UDR] - Dream Router
[UDRULTRA] - Dream Router Ultra
[UDW] - Dream Wall
[UCK] - Cloud Key
[UCKG2] - Cloud Key G2 Plus
[USG] - Secure Gateway
[USGPRO] - Secure Gateway Professional
[UCKENT] - Cloud Key Enterprise
[UCC] - UniFi Cloud Console - For cloud consoles, just use [UCC]-[COMPANYNAME], as these consoles don't have a specific location

Remaining UniFi devices should be named as appropriate for their location or function. The primary root switch should be named "Core Switch", while APs should generally be labeled with a specific location such as "Conference Room AP" or "Room 3118 AP"

Routers/Gateways/Firewalls

Thursday, November 2, 2023 1:31 PM

Set up primary Wi-Fi network using WPA3 only and all bands

Set up guest network (if requested) using WPA2 or WPA2/3 hybrid mode, all APs, and the 2.4/5GHz bands only. The 6 GHz performance band should be limited to the primary network only

Create a speed limit profile in the UDM under Settings - Profiles - Wi-Fi speed limit. Generally, limit this to 25% each of the maximum upload and download bandwidth of the incoming WAN connection

Configure a captive portal for the guest network (if requested), enable HTTPS, enforce branding, enforce terms of service, and redirect the success landing page to the client's main webpage.

Enforce the speed limits configured above on the guest network by going to Settings -> WiFi -> [guest network name] -> WiFi Speed Limit and selecting the guest speed limit profile.

Under Settings -> WiFi -> [guest network name] -> WiFi Scheduler, create a time schedule to turn off guest WiFi outside of the business' main hours of operation to reduce the risk of incidental attacks. As a typical guideline, you can shut off the guest network from 20:00 to 04:00 daily.

Under Settings -> Teleport and VPN, under the Teleport tab, turn off the Teleport VPN. The Teleport app only works on iOS, Android, and macOS.

Under Settings -> Application Firewall, set the Suspicious Activity toggle to Notify and Block, and set Detection Sensitivity to High. Also check the Dark Web Blocker and Malicious Website Blocker.

Under Settings -> Application Firewall -> Gateway Engine, turn on Device and Traffic identification, as well as Country Restrictions. Block all countries indicated in the Country Block List on this page.

Under Settings -> System -> Updates, uncheck Automate Device Updates and add a new device update schedule that repeats monthly at 00:00 hours. Select all applicable devices. This will allow a sufficient amount of time for updates to exist in the wild and be replaced if they break functionality before they apply to the primary network, but also keep the network relatively up to date when it comes to security updates.

Under Settings -> System -> Backups, click on the "Console Cloud Backups" hyperlink at the bottom of the page to go to the cloud backup manager. In the new Console Settings window that opens up, make sure System Config Backup is checked and that a recent backup has been performed.

Under Settings -> System -> Advanced, make sure the WiFiman Support box is checked so that the WiFiman app can be used to calibrate device installation location and signal strength.

In Settings -> System -> Advanced, under Wireless Connectivity, disable Wireless Meshing if all access points in the site are hard-wired to the controller. If you're using wireless UniFi devices, such as the wireless G4 Doorbell or UniFi Protect wireless cameras, leave this setting enabled.

COUNTRY BLOCK LIST

- Afghanistan
- Azerbaijan
- Bangladesh
- Bahrain
- Brazil
- Belarus
- Congo [DRC]
- Central African Republic
- Congo [ROC]
- China
- Colombia
- Cuba
- Egypt
- Hong Kong
- Honduras
- Indonesia
- India
- Iraq
- Iran
- Jordan
- Kyrgyzstan
- North Korea
- Kazakhstan
- Sri Lanka
- Libya
- Myanmar
- Mongolia
- Nigeria
- Oman
- Philippines
- Pakistan
- Qatar
- Russia
- Rwanda
- Saudi Arabia
- Sudan
- Somalia
- South Africa
- Syria
- Thailand
- Turkmenistan
- Turkey
- United Arab Emirates
- Uzbekistan
- Venezuela
- Vietnam
- Yemen
- Zimbabwe

Switches

Thursday, November 2, 2023 1:31 PM

Set the priority of switches in UniFi Devices -> Switch in question -> Settings tab.

Proper switching architecture dictates that the main router is only connected to the core switch - do not connect any devices to the UDM/UDR/UDW except the primary core switch.

Set the primary core switch RTSP priority in switch settings -> Advanced to 4096. This should be the ONLY switch assigned this priority at each site. For each switch behind the core switch, adjust the priority up one step for each layer of separation between that switch and the core switch. For example, the core switch will have priority 4096. Any secondary switches connected to the core switch directly will ALL have priority 8192. Any tertiary switches connected directly to the secondary switches will ALL have priority 12288, and so on and so forth.

Please note: this advice given above is a recommendation for setups with one core switch. For setups where you believe the existing architecture may be placed under a new higher-level switch down the line (such as an aggregation or leaf switch), start your core switch priority at 8192 and go from there, reserving 4096 for a future aggregator switch.

Any core switch connected directly to a UniFi Gateway should be connected using the highest-bandwidth port available. For smaller switches, this is usually an RJ45 port, but for more advanced switches a 1G or 10G SFP-to-SFP "direct attach cable" should be used to maximize available bandwidth on the RJ45 backplane. This offloads routing between the core switch and main gateway to the SFP ports, which are on their own backplane that does not share bandwidth with the RJ45 port backplane. A sample 1G direct attach cable looks like this:

https://www.amazon.com/dp/B09P517YVM/

Access Points

Thursday, November 2, 2023 1:31 PM

Update all APs to latest version

If all APs are hardwired, go into device settings for each AP and disable meshing. This setting is controlled per AP, not in the Global AP Settings

Under Settings -> Global AP Settings in the UDM, set transmit power to Medium for all radios. This can be adjusted after installation.

For the 2.4 GHz band, always set this to the lowest channel width to maximize coverage on the fallback radio

For the 5.0 GHz band, set this to 40 MHz. If you have uninterrupted line of sight to >80% of client devices from the AP's final installation location, you can set this to 80 MHz

For the 6 GHz band, set this to 160 MHz. This will maximize bandwidth for supported clients within range.

WAN Failover Devices

Thursday, November 2, 2023 1:31 PM

Raise data limit to 5GB/10GB in device settings - default is to shut off after 3GB

Enable external antenna support in device settings ONLY AFTER CONNECTING THE EXTERNAL RP-SMA antenna

Connect the U-LTE to the backup WAN port - if primary WAN port is using RJ45, you'll need a 1G or 10G/1G auto negotiating SFP+ to RJ45 module like this one from Ubiquiti: https://store.ui.com/us/en/pro/category/accessories-cables-dacs/products/rj45-to-sfp-module?variant=UACC-CM-RJ45-1G